Data Analytics using Python

Week 1 Question Bank Answer

1 Define data and its importance

Ans.1) Data is a collection of facts, figures, or information that can be processed, analyzed, and used to make decisions, predictions, or conclusions. It can be generated by humans, machines, or a combination of both and can be stored in structured or unstructured formats. Data is important because it helps in making better decisions, solving problems, evaluating performance, improving processes, understanding consumers and the market, and adding value to businesses.

Data analytics is the scientific process of transforming data into insights for making better decisions. It is the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions. Data analysis is the process of examining, transforming, and arranging raw data in a specific way to generate useful information from it. It allows for the evaluation of data through analytical and logical reasoning to lead to some sort of outcome or conclusion in some context.

Data analytics is important because it can help organizations to determine credit risk, develop new medicines, find more efficient ways to deliver products and services, prevent fraud, uncover cyber threats, retain the most valuable customers, and make better decisions based on data-driven insights. It is a multi-faceted process that involves a number of steps, approaches, and diverse techniques, and can be classified into four major types: descriptive analytics, diagnostic analytics, predictive analytics, and prescriptive analytics.

Descriptive analytics is the conventional form of Business Intelligence and data analysis that seeks to provide a depiction or “summary view” of facts and figures in an understandable format. It can summarize raw data and convert it into a form that can be easily understood by humans. Diagnostic analytics is a form of advanced analytics that examines data or content to answer the question “Why did it happen?” Predictive analytics helps to forecast trends based on the current events, and prescriptive analytics indicates the best course of action to optimize the outcome. These types of data analytics can help organizations to improve quality, enhance services, reduce costs, increase productivity, and make better decisions based on data-driven insights.

2 Define data analytics and its types

Ans.2) Data analytics is the scientific process of transforming data into insights for making better decisions. It involves the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions.

There are four major types of data analytics:

1. **Descriptive analytics**: This is the conventional form of Business Intelligence and data analysis. It seeks to provide a depiction or “summary view” of facts and figures in an understandable format. It can summarize raw data and convert it into a form that can be easily understood by humans. Descriptive analysis or statistics can describe in detail about an event that has occurred in the past.
2. **Diagnostic analytics**: This is a form of advanced analytics which examines data or content to answer the question “Why did it happen?” Diagnostic analytical tools aid an analyst to dig deeper into an issue so that they can arrive at the source of a problem.
3. **Predictive analytics**: This helps to forecast trends based on the current events. Predicting the probability of an event happening in future or estimating the accurate time it will happen can all be determined with the help of predictive analytical models.
4. **Prescriptive analytics**: This indicates the best course of action to optimize the outcome. The goal of prescriptive analytics is to enable quality improvements, service enhancements, cost reductions, and increasing productivity.

Data analytics is important because it can help organizations to determine credit risk, develop new medicines, find more efficient ways to deliver products and services, prevent fraud, uncover cyber threats, retain the most valuable customers, and make better decisions based on data-driven insights.

3 Explain why analytics is important in today's business environment.

Ans.3) Analytics is important in today's business environment for several reasons:

- **Demand for Data Analytics**: The demand for data analytics has been increasing rapidly in recent years. According to a report by Times of India, data scientists are earning more than chartered accountants and engineers in India. This is because businesses are recognizing the value of data analytics in making informed decisions and gaining a competitive advantage.
- **Element of Data Analytics**: Analytics is an essential element of data-driven decision-making. It involves the use of data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models to help managers gain improved insight about their business operations and make better, fact-based decisions.
- **Data-Driven Decision Making**: Analytics helps businesses to make data-driven decisions by providing insights into customer behavior, market trends, and operational efficiency. By analyzing data, businesses can identify patterns and trends that would otherwise be difficult to detect.
- **Predictive Analytics**: Predictive analytics helps businesses to forecast trends based on current events. This can help businesses to anticipate future outcomes and make informed decisions.
- **Prescriptive Analytics**: Prescriptive analytics indicates the best course of action to optimize the outcome. The goal of prescriptive analytics is to enable quality improvements, service enhancements, cost reductions, and increasing productivity.
- **Data Visualization**: Analytics can help businesses to visualize data in a way that is easy to understand and interpret. This can help businesses to identify trends and patterns that would otherwise be difficult to detect.
- **Data-Driven Culture**: Analytics can help businesses to create a data-driven culture. By using data to inform decision-making, businesses can make more informed decisions that are based on facts rather than intuition.

In summary, analytics is important in today's business environment because it helps businesses to make data-driven decisions, forecast trends, optimize outcomes, visualize data, and create a data-driven culture. By using analytics, businesses can gain a competitive advantage, improve operational efficiency, and make informed decisions that are based on facts rather than intuition.

4 Explain how statistics, analytics and data science are interrelated

Ans.4) Statistics, analytics, and data science are interrelated fields that share a common focus on extracting insights from data.

Statistics is a branch of mathematics that deals with the collection, analysis, interpretation, and presentation of data. It provides tools and techniques for summarizing and describing data, as well as for making inferences and predictions based on data.

Analytics is the process of examining data to extract insights and make informed decisions. It involves using statistical techniques and other tools to analyze data and identify patterns, trends, and relationships.

Data science is a multidisciplinary field that combines elements of statistics, computer science, and domain expertise to extract insights from large and complex data sets. It involves using statistical techniques, machine learning algorithms, and other tools to analyze data and make predictions or recommendations.

In summary, statistics provides the foundational mathematical tools for analyzing data, analytics is the process of applying these tools to extract insights, and data science is a multidisciplinary field that combines elements of both to tackle complex data analysis problems.

5 Why python?

Ans.5) Python is a popular programming language for data analytics due to its simplicity, ease of use, and extensive libraries. Here are some reasons why Python is a good choice for data analytics:

1. **Simple and easy to learn**: Python has a simple syntax and is easy to learn, making it a good choice for beginners.
2. **Freeware and open source**: Python is free to use and open source, which means that there are no licensing fees and you can customize it to meet your needs.
3. **Interpreted**: Python is an interpreted language, which means that you can run code immediately without the need for a separate compilation step.
4. **Dynamically typed**: Python is dynamically typed, which means that you don't need to specify the data type of a variable when you declare it.
5. **Extensible**: Python is extensible, which means that you can add new functionality to the language by writing extensions in C or C++.
6. **Embedded**: Python can be embedded in other applications, which means that you can use it to add scripting capabilities to your existing software.
7. **Extensive library**: Python has a large and active community, which has created a vast library of modules and tools for data analytics. Some popular libraries for data analytics in Python include NumPy, Pandas, Matplotlib, and Scikit-learn.

Python is a versatile language that can be used for a wide range of data analytics tasks, from simple data manipulation and visualization to advanced machine learning and artificial intelligence. It is a good choice for both beginners and experienced data analysts.

6 Explain the four different levels of Data

Ans.6) The four different levels of data measurement are nominal, ordinal, interval, and ratio.

1. **Nominal**: Nominal data is the lowest level of measurement and is used to classify data into distinct categories with no ranking or order. Examples of nominal data include gender, marital status, and political party affiliation.
2. **Ordinal**: Ordinal data is similar to nominal data, but it includes a ranking or order. Examples of ordinal data include product satisfaction (satisfied, neutral, unsatisfied), faculty rank (professor, associate professor, assistant professor), and student grades (A, B, C, D, F).
3. **Interval**: Interval data is an ordered scale in which the difference between measurements is a meaningful quantity, but the measurements do not have a true zero point. Examples of interval data include temperature in Fahrenheit or Celsius and year.
4. **Ratio**: Ratio data is the highest level of measurement and is an ordered scale in which the difference between the measurements is a meaningful quantity and the measurements have a true zero point. Examples of ratio data include weight, age, and salary.

The choice of measurement scale impacts the meaningful operations that can be performed on the data and the statistical methods that can be used. Nominal data can only be classified and counted, while ordinal data can also be ranked. Interval data can be added, subtracted, and compared, while ratio data can be multiplied and divided in addition to the operations that can be performed on interval data. Nonparametric statistical methods are used for nominal and ordinal data, while parametric methods are used for interval and ratio data.

7 Define Variable, Measurement and Data

Ans.7) Variable, measurement, and data are all related concepts in the field of data analytics.

- **Variable**: A variable is a characteristic of any entity being studied that is capable of taking on different values. For example, the number of children a person has, the weight of a product, or the temperature in a room.
- **Measurement**: Measurement is the process of assigning numbers to particular attributes or characteristics of a variable. This involves using a standard process to assign numerical values to the variable. For example, counting the number of children a person has, weighing a product, or using a thermometer to measure the temperature in a room.
- **Data**: Data are recorded measurements. Once measurements have been taken, they can be recorded and analyzed to extract insights. For example, data on the number of children a person has can be used to analyze trends in family size, data on the weight of a product can be used to optimize packaging and shipping, and data on room temperature can be used to maintain a comfortable environment.

In summary, variable, measurement, and data are all interrelated concepts that are used to collect, record, and analyze information in data analytics. A variable is a characteristic that can take on different values, measurement is the process of assigning numbers to those values, and data are the recorded measurements that can be analyzed to extract insights.

8 **Explain Data analytics vs. Data analysis.**

Ans.8)

Criteria	Data Analytics	Data Analysis
Definition	The scientific process of transforming data into insights for making better decisions	The process of examining, transforming, and arranging raw data in a specific way to generate useful information from it
Focus	Transforming data into actionable insights to drive decision-making and improve business outcomes	Examining and organizing data to derive useful information
Methodology	Uses data, information technology, statistical analysis, quantitative methods, and mathematical or computer-based models	Examines data through analytical and logical reasoning
Scope	Broader in scope, encompassing the entire process of deriving insights from data	A specific step within the broader data analytics process
Types	Descriptive analytics, diagnostic analytics, predictive analytics, prescriptive analytics	Data analysis, data mining, data discovery, data modeling
Purpose	To help managers gain improved insight about their business operations and make better, fact-based decisions	To evaluate data through analytical and logical reasoning to lead to some sort of outcome or conclusion in a given context
Techniques	Regression analysis, time series analysis, machine learning, data mining, statistical modeling	Data visualization, data aggregation, data transformation, data mining
Outcome	Actionable insights, recommendations, and decisions	Reports, dashboards, visualizations, and summaries
Example	Predicting customer churn, identifying fraudulent transactions, optimizing supply chain operations	Sales reports, customer segmentation, website analytics, social media analytics

Data Analytics and Data Analysis are both important concepts in the field of data science, but they have different focuses and methodologies.

Data Analytics is a broader concept that encompasses the entire process of deriving insights from data, while Data Analysis is a specific step within this process that involves examining and organizing data to derive useful information.

Data Analytics uses various techniques such as regression analysis, time series analysis, machine learning, and statistical modeling to transform data into actionable insights, while Data Analysis uses data visualization, data aggregation, data transformation, and data mining techniques to examine and organize data.

The outcome of Data Analytics is actionable insights, recommendations, and decisions, while the outcome of Data Analysis is reports, dashboards, visualizations, and summaries.

9 Give classification of Data Analytics.

Ans.9) Data Analytics can be classified into four major types based on the phase of workflow and the kind of analysis required:

1. **Descriptive analytics**: This is the conventional form of Business Intelligence and data analysis. It seeks to provide a depiction or "summary view" of facts and figures in an understandable format. This either informs or prepares data for further analysis. Descriptive analysis or statistics can summarize raw data and convert it into a form that can be easily understood by humans. They can describe in detail about an event that has occurred in the past.
2. **Diagnostic analytics**: This is a form of advanced analytics which examines data or content to answer the question "Why did it happen?" Diagnostic analytical tools aid an analyst to dig deeper into an issue so that they can arrive at the source of a problem.
3. **Predictive analytics**: Predictive analytics helps to forecast trends based on the current events. Predicting the probability of an event happening in future or estimating the accurate time it will happen can all be determined with the help of predictive analytical models.
4. **Prescriptive analytics**: This indicates the best course of action to optimize the outcome. The goal of prescriptive analytics is to enable quality improvements, service enhancements, cost reductions, and increasing productivity.

These types of data analytics can help organizations to improve quality, enhance services, reduce costs, increase productivity, and make better decisions based on data-driven insights.

10 Give different Types of Variables.

Ans.10) Variables are a fundamental concept in data analytics and can be classified into different types based on their characteristics and measurement levels. Here are the different types of variables:

1. **Categorical variables**: These are variables that can be grouped into categories or classes. They can be further classified into two types:

- **Nominal variables**: These are categorical variables that do not have a natural order or ranking. Examples include gender, eye color, and political party.
- **Ordinal variables**: These are categorical variables that have a natural order or ranking. Examples include product satisfaction (satisfied, neutral, unsatisfied), faculty rank (professor, associate professor, assistant professor), and student grades (A, B, C, D, F).

2. **Numeric variables**: These are variables that can be measured and expressed in numerical values. They can be further classified into two types:

- **Interval variables**: These are numeric variables that have a fixed unit of measurement and an arbitrary zero point. Examples include temperature in Fahrenheit or Celsius and year.
- **Ratio variables**: These are numeric variables that have a fixed unit of measurement and a true zero point. Examples include weight, age, and salary.

The choice of measurement scale impacts the meaningful operations that can be performed on the data and the statistical methods that can be used. Nominal data can only be classified and counted, while ordinal data can also be ranked. Interval data can be added, subtracted, and compared, while ratio data can be multiplied and divided in addition to the operations that can be performed on interval data. Nonparametric statistical methods are used for nominal and ordinal data, while parametric methods are used for interval and ratio data.

In summary, variables can be classified into different types based on their characteristics and measurement levels, including categorical and numeric variables, and further classified into nominal, ordinal, interval, and ratio variables. The choice of measurement scale impacts the meaningful operations that can be performed on the data and the statistical methods that can be used.

11 Give the usage Potential of Various Levels of Data.

Ans.11) The potential usage of various levels of data is as follows:

1. **Nominal**: Nominal data is the lowest level of measurement and is used to classify data into distinct categories in which no ranking is implied. Nominal data can only be classified and counted, and nonparametric statistical methods are used for analysis. Examples of nominal data include gender, marital status, and political party affiliation.
2. **Ordinal**: Ordinal data is similar to nominal data, but it includes a ranking or order. Ordinal data can be classified, counted, and ranked, and nonparametric statistical methods are used for analysis. Examples of ordinal data include product satisfaction, faculty rank, and student grades.
3. **Interval**: Interval data is an ordered scale in which the difference between measurements is a meaningful quantity, but the measurements do not have a true zero point. Interval data can be added, subtracted, and compared, and parametric statistical methods are used for analysis. Examples of interval data include temperature in Fahrenheit or Celsius and year.
4. **Ratio**: Ratio data is the highest level of measurement and is an ordered scale in which the difference between the measurements is a meaningful quantity and the measurements have a true zero point. Ratio data can be added, subtracted, multiplied, and divided, and parametric statistical methods are used for analysis. Examples of ratio data include weight, age, and salary.

In summary, the potential usage of various levels of data depends on the level of measurement and the type of statistical analysis required. Nominal data is used for classification and counting, ordinal data is used for classification, counting, and ranking, interval data is used for addition, subtraction, and comparison, and ratio data is used for addition, subtraction, multiplication, and division. Nonparametric statistical methods are used for nominal and ordinal data, while parametric methods are used for interval and ratio data.

12 Explain following methods:

Head(),Get the number of rows and columns

get column names, get the dtype of each column

get more information about data

Subset Rows by Index Label: loc

get the first row

get the last row

Subsetting Multiple Rows

Subset Rows by Row Number: iloc

Subsetting Columns

Subsetting Columns by Range

Subsetting Rows and Columns

Subsetting Multiple Rows and Columns

Calculating Grouped Means

Grouped Frequency Counts

Ans.12) The following methods are commonly used in data analysis to manipulate and examine data:

1. **Head()**: This method returns the first n rows of a DataFrame, with the default value of n being 5. It is used to quickly preview the contents of a dataset.
2. **Get the number of rows and columns**: This can be done using the shape attribute of a DataFrame, which returns a tuple with the number of rows and columns.
3. **Get column names**: This can be done using the columns attribute of a DataFrame, which returns a list of column names.
4. **Get the dtype of each column**: This can be done using the dtypes attribute of a DataFrame, which returns a Series with the data types of each column.
5. **Get more information about data**: This can be done using the info() method of a DataFrame, which returns a summary of the data including the number of non-null values, data types, and memory usage.
6. **Subset Rows by Index Label: loc**: This method is used to select rows based on their index labels.

   For example, df.loc[label] returns the row with the specified label.
7. **Get the first row**: This can be done using the iloc method, which returns the first row of the DataFrame.
8. **Get the last row**: This can be done using the iloc[-1] method, which returns the last row of the DataFrame.
9. **Subsetting Multiple Rows**: This can be done using the loc or iloc methods with a list of labels or indices.

   For example, df.loc[[label1, label2]] returns the rows with the specified labels.
10. **Subset Rows by Row Number: iloc**: This method is used to select rows based on their row numbers.

    For example, df.iloc[row_number] returns the row with the specified number.
11. **Subsetting Columns**: This can be done using the loc or iloc methods with a list of column labels or indices.

    For example, df.loc[:, column_labels] returns the columns with the specified labels.
12. **Subsetting Columns by Range**: This can be done using the iloc method with a slice notation.

    For example, df.iloc[:, 1:3] returns the columns from the second to the third column.
13. **Subsetting Rows and Columns**: This can be done using the loc or iloc methods with a list of row labels or indices and a list of column labels or indices.

    For example, df.loc[[label1, label2], column_labels] returns the rows and columns with the specified labels.
14. **Subsetting Multiple Rows and Columns**: This can be done using the loc or iloc methods with a boolean mask.

    For example, df.loc[df['column_name'] > value, column_labels] returns the rows and columns where the specified condition is true.
15. **Calculating Grouped Means**: This can be done using the groupby method followed by an aggregation function such as mean.

    For example, df.groupby('column_name')['target_column'].mean() returns the mean of the target column for each group.
16. **Grouped Frequency Counts**: This can be done using the value_counts method.

    For example, df['column_name'].value_counts() returns the frequency counts of each unique value in the specified column.

These methods are useful for exploring and manipulating data in a DataFrame, which is a common data structure used in data analysis.

13 Explain Pandas Types Versus Python Types

Ans.13)

Pandas Types	Python Types	Description
object	str	Most common data type
int64	int	Whole numbers
float64	float	Numbers with decimals
bool	bool	Boolean values (True or False)
datetime64	datetime	datetime is found in the Python standard library

In Pandas, the data types are slightly different from the standard Python data types. Pandas introduces specific data types to handle data more efficiently and accurately. The table above shows the correspondence between Pandas types and Python types.

14 Give various methods of Visual Representation of the Data.

Ans.14) Visual representation of data is an essential aspect of data analytics as it helps to convey complex data in an easy-to-understand format. Various methods of visual representation of data are as follows:

1. **Histogram**: A histogram is a vertical bar chart of frequencies. It displays the distribution of a dataset by dividing it into intervals or bins and showing the number of observations that fall within each bin.
2. **Frequency Polygon**: A frequency polygon is a line graph of frequencies. It is created by connecting the midpoints of the tops of the bars in a histogram.
3. **Ogive**: An ogive is a line graph of cumulative frequencies. It shows the cumulative frequency of a dataset as a function of the variable being measured.
4. **Pie Chart**: A pie chart is a proportional representation for categories of a whole. It shows the proportion of each category in a dataset as a slice of a pie.
5. **Stem and Leaf Plot**: A stem and leaf plot is a graphical representation of a dataset that displays the distribution of the data and the values of individual observations.
6. **Pareto Chart**: A Pareto chart is a combination of a bar chart and a line graph. It shows the relative frequency of different categories in a dataset and the cumulative total of those categories.
7. **Scatter Plot**: A scatter plot is a graph that displays the relationship between two variables. It shows the values of one variable on the x-axis and the values of the other variable on the y-axis.
8. **Table**: A table is a simple and effective way to display data in a structured format. It can be used to display summary statistics, such as the mean, median, and standard deviation, or to display the raw data itself.
9. **Graphs**: Graphs are a powerful tool for visualizing data. They can be used to display trends, patterns, and relationships in data.
10. **Pie Chart**: A pie chart is a circular chart divided into sectors, each representing a category or proportion of the whole.
11. **Multiple Bar Chart**: A multiple bar chart is a chart that displays multiple bars for each category, allowing for easy comparison between categories.
12. **Simple Pictogram**: A simple pictogram is a chart that uses pictures or icons to represent data.
13. **Multiple Line Chart**: A multiple line chart is a chart that displays multiple lines for each category, allowing for easy comparison between categories over time.
14. **Area Chart**: An area chart is a chart that displays the cumulative total of a dataset over time, with the area below the line filled in to represent the total.
15. **Bubble Chart**: A bubble chart is a chart that displays data points as bubbles, with the size and color of the bubbles representing different variables.
16. **Heat Map**: A heat map is a chart that displays data as colors, with darker colors representing higher values and lighter colors representing lower values.
17. **Radar Chart**: A radar chart is a chart that displays data as points on a radar-like diagram, allowing for easy comparison between categories.
18. **Waterfall Chart**: A waterfall chart is a chart that displays the cumulative effect of a series of positive and negative values, showing how they contribute to the final value.
19. **Funnel Chart**: A funnel chart is a chart that displays the progression of a dataset through a series of stages, with each stage represented as a narrower section of the funnel.
20. **Gauge Chart**: A gauge chart is a chart that displays data as a dial or gauge, with the position of the dial representing the value of the data.

These methods of visual representation of data can be used to convey complex data in an easy-to-understand format, helping to facilitate decision-making and communication.

15 Give Measures of Central Tendency

Ans.15) Measures of central tendency are statistical measures that provide information about the central location or typical value of a dataset. There are three main measures of central tendency:

1. **Mean**: The mean is the average of a dataset and is calculated by summing all the values in the dataset and dividing the sum by the number of values in the dataset. It is applicable for interval and ratio data and is affected by each value in the dataset, including extreme values.

   Example: For the dataset {2, 5, 7, 9, 11}, the mean is (2 + 5 + 7 + 9 + 11) / 5 = 7.4.

2. **Median**: The median is the middle value in an ordered array of numbers and is applicable for ordinal, interval, and ratio data. It is unaffected by extremely large and extremely small values. The median's position in an ordered array is given by (n+1)/2, where n is the number of terms in the dataset.

   Example: For the dataset {2, 5, 7, 9, 11}, the median is 7.

3. **Mode**: The mode is the most frequently occurring value in a dataset and is applicable to all levels of data measurement, including nominal, ordinal, interval, and ratio data. A dataset can have more than one mode, in which case it is called bimodal or multimodal.

   Example: For the dataset {2, 5, 7, 7, 9, 11}, the mode is 7.

In summary, the mean is the average of all the values in the dataset, the median is the middle value in an ordered array of numbers, and the mode is the most frequently occurring value in a dataset.

These measures of central tendency are used to summarize and describe the characteristics of a dataset and are used in various applications, including data analysis, statistics, and machine learning.

16 How to calculate mean, mode and median of ungrouped and grouped data?

Ans.16) Measures of central tendency are statistical measures that provide information about the center or typical value of a dataset. There are three main measures of central tendency: mean, median, and mode.

To calculate the mean, median, and mode of ungrouped and grouped data, we can use the following formulas:

1. Mean: The mean is the sum of all the values in the dataset divided by the number of values.

For ungrouped data, the formula is:

Mean = (x1 + x2 + ... + xn) / n

where x1, x2, ..., xn are the values in the dataset and n is the number of values.

For grouped data, the formula is:

Mean = (f1 * M1 + f2 * M2 + ... + fn * Mn) / (f1 + f2 + ... + fn)

where f1, f2, ..., fn are the frequencies of the class intervals and M1, M2, ..., Mn are the class midpoints.

2. Median: The median is the middle value in an ordered dataset. For odd-numbered datasets, the median is the value at the (n+1)/2 index, where n is the number of values. For even-numbered datasets, the median is the average of the values at the (n/2) and (n/2+1) indices.

For grouped data, the formula is:

Median = L + (n/2 - cf) * w

where L is the lower limit of the median class, n is the number of values, cf is the cumulative frequency of the class preceding the median class, and w is the width of the median class.

3. Mode: The mode is the value that occurs most frequently in a dataset. For ungrouped data, the mode is the value that occurs most frequently. For grouped data, the mode is the class interval with the highest frequency.

Example:

Suppose we have the following ungrouped dataset:

3, 6, 7, 8, 9, 10, 11, 12, 15

To calculate the mean, we add up all the values and divide by the number of values:

Mean = (3 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 15) / 9 = 8.33

To calculate the median, we first order the dataset:

3, 6, 7, 8, 9, 10, 11, 12, 15

Since there are 9 values, the median is the value at the (9+1)/2 index, which is the 5th value:

Median = 9

To calculate the mode, we look for the value that occurs most frequently. In this case, there is no single value that occurs more than once, so there is **no mode**.

For grouped data, suppose we have the following frequency distribution:

Class Interval	Frequency(f)
0-5	10
5-10	15
10-15	20
15-20	12
20-25	8

To calculate the mean, we first find the class midpoints:

Class Interval	Frequency(f)	Class Midpoint(M)
0-5	10	2.5
5-10	15	7.5
10-15	20	12.5
15-20	12	17.5
20-25	8	22.5

Then we use the formula:

Mean = (10 * 2.5 + 15 * 7.5 + 20 * 12.5 + 12 * 17.5 + 8 * 22.5) / (10 + 15 + 20 + 12 + 8) = 12.5

To calculate the median, we first find the cumulative frequency:

Class Interval	Frequency(f)	Cumulative Frequency(cf)
0-5	10	10
5-10	15	25
10-15	20	45
15-20	12	57
20-25	8	65

Then we use the formula:

Median = 10 + (65/2 - 45) * (5 - 10) / (15 - 10) = 12.5

To calculate the mode, we look for the class interval with the highest frequency:

Class Interval	Frequency(f)
0-5	10
5-10	15
10-15	20
15-20	12
20-25	8

The class interval with the highest frequency is 10-15, so the mode is 12.5.

In summary, the mean, median, and mode are measures of central tendency that provide information about the center of a dataset. The mean is the sum of all the values divided by the number of values, the median is the middle value in an ordered dataset, and the mode is the value that occurs most frequently. For grouped data, the mean is calculated using the weighted average of class midpoints, the median is calculated using the formula L + (n/2 - cf) * w, and the mode is the class interval with the highest frequency.

17 Give the Measures of Variability or dispersion.

Ans.17) Measures of variability or dispersion are statistical measures that provide information about the spread or dispersion of a set of data. There are several common measures of variability, including:

1. **Range**: The difference between the largest and smallest values in a set of data. It is simple to compute, but it ignores all data points except the two extremes.
2. **Inter-quartile range (IQR)**: The difference between the 75th percentile and the 25th percentile of a set of data. It is a measure of the spread of the middle 50% of the data.
3. **Mean Absolute Deviation (MAD)**: The average of the absolute deviations from the mean. It is a measure of the average distance between each data point and the mean.
4. **Variance**: The average of the squared deviations from the mean. It is a measure of the spread of the data around the mean.
5. **Standard Deviation (SD)**: The square root of the variance. It is a measure of the spread of the data around the mean, expressed in the same units as the data.
6. **Z scores**: The number of standard deviations that a data point is above or below the mean. It is a measure of how many standard deviations a data point is from the mean.
7. **Coefficient of Variation (CV)**: The ratio of the standard deviation to the mean, expressed as a percentage. It is a measure of the relative variability of the data.

These measures of variability provide information about the spread or dispersion of a set of data, which is important for understanding the reliability of measures of central tendency and for comparing the dispersion of various samples.

For example, consider the following set of data: 35, 37, 37, 39, 40, 40, 41, 41, 43, 43, 43, 43, 44, 44, 44, 44, 44, 45, 45, 46, 46, 46, 46.

The mean of the data is 42.45, the median is 43, and the mode is 44.

The range of the data is 46 - 35 = 11, the inter-quartile range is 46 - 41 = 5, the mean absolute deviation is 3.02, the variance is 16.81, and the standard deviation is 4.10.

These measures of variability provide information about the spread or dispersion of the data, which is important for understanding the reliability of measures of central tendency and for comparing the dispersion of various samples.

In summary, measures of variability or dispersion are statistical measures that provide information about the spread or dispersion of a set of data. There are several common measures of variability, including range, inter-quartile range, mean absolute deviation, variance, standard deviation, z scores, and coefficient of variation. These measures are important for understanding the reliability of measures of central tendency and for comparing the dispersion of various samples.

18 **Define Common Measures of Variability**

i) **Range**
ii) **Inter-quartile range**
iii) **Mean Absolute Deviation**
iv) **Variance**
v) **Standard Deviation**
vi) **Z scores**
vii) **Coefficient of Variation**

Ans.18) Common measures of variability include:

1. **Range**: The difference between the largest and smallest values in a dataset.
2. **Inter-quartile range (IQR)**: The difference between the 75th percentile and the 25th percentile of a dataset.
3. **Mean Absolute Deviation (MAD)**: The average of the absolute deviations from the mean.
4. **Variance**: The average of the squared deviations from the mean.
5. **Standard Deviation (SD)**: The square root of the variance.
6. **Z scores**: The number of standard deviations that a data point is above or below the mean.
7. **Coefficient of Variation (CV)**: The ratio of the standard deviation to the mean, expressed as a percentage.

The range provides a simple measure of the spread of a dataset, while the IQR is a more robust measure that is less affected by outliers.

The MAD, variance, and standard deviation provide measures of the dispersion of a dataset around the mean, with the standard deviation being the most commonly used measure.

Z scores and the coefficient of variation provide measures of the dispersion of a dataset relative to the mean, allowing for comparisons between datasets with different means and standard deviations.

These measures of variability are important for understanding the dispersion of a dataset and for identifying any patterns or trends in the data.
By comparing measures of variability across different datasets, it is possible to identify differences in the dispersion of the data and to make more informed decisions based on the data.

OR

Measures of variability, also known as measures of dispersion, describe the spread or dispersion of a dataset. Here are the definitions and examples of the common measures of variability:

i) **Range**: The range is the difference between the largest and smallest values in a dataset. It is a simple measure of variability that provides a quick overview of the spread of the data.

Example: The range of the dataset {2, 4, 6, 8, 10} is 10 - 2 = 8.

ii) **Inter-quartile range (IQR)**: The inter-quartile range is the difference between the 75th percentile (Q3) and the 25th percentile (Q1) of a dataset. It is a measure of the spread of the middle 50% of the data and is less affected by outliers than the range.

Example: The IQR of the dataset {2, 4, 6, 8, 10, 12, 14, 16, 18, 20} is 18 - 6 = 12.

iii) **Mean Absolute Deviation (MAD)**: The mean absolute deviation is the average of the absolute differences between each value in the dataset and the mean. It is a measure of the typical distance of the data points from the mean.

Example: The MAD of the dataset {2, 4, 6, 8, 10} is (|2 - 6| + |4 - 6| + |6 - 6| + |8 - 6| + |10 - 6|) / 5 = 3.2.

iv) **Variance**: Variance is the average of the squared deviations from the mean. It quantifies the spread of data points around the mean, with higher variance indicating greater dispersion.

Example: The variance of the dataset {2, 4, 6, 8, 10} is ($|2 - 6|^2$ + $|4 - 6|^2$ + $|6 - 6|^2$ + $|8 - 6|^2$ + $|10 - 6|^2$) / 5 = 10.4.

v) **Standard Deviation (SD)**: The standard deviation is the square root of the variance. It provides a measure of the dispersion of data points around the mean in the same units as the data, making it easier to interpret.

Example: The SD of the dataset {2, 4, 6, 8, 10} is the square root of 10.4, which is 3.22.

vi) **Z scores**: Z scores indicate how many standard deviations a data point is above or below the mean. They standardize data, allowing for comparisons between datasets with different means and standard deviations.

Example: The Z score of the value 4 in the dataset {2, 4, 6, 8, 10} is (4 - 6) / 3.22 = -0.62.

vii) **Coefficient of Variation (CV)**: The coefficient of variation is the ratio of the standard deviation to the mean, expressed as a percentage. It provides a relative measure of variability, allowing for comparisons between datasets with different means and standard deviations.

Example: The CV of the dataset {2, 4, 6, 8, 10} is (3.22 / 6) x 100% = 53.67%.

19 What are Measures of Shape?

Ans.19) Measures of shape are statistical measures that provide information about the distribution or shape of a dataset. These measures help to describe the form of the data distribution and provide insights into the symmetry, skewness, and kurtosis of the data. Common measures of shape include:

1. **Skewness**: Skewness is a measure of the asymmetry of the distribution of data. It indicates whether the data is skewed to the left or right. A skewness value of 0 indicates a symmetrical distribution, while positive skewness indicates a right-skewed distribution and negative skewness indicates a left-skewed distribution.
2. **Kurtosis**: Kurtosis is a measure of the peakedness or flatness of the distribution of data. It indicates how much data is concentrated around the mean. A kurtosis value of 3 indicates a normal distribution, while values greater than 3 indicate a peaked distribution (leptokurtic) and values less than 3 indicate a flat distribution (platykurtic).

Measures of shape provide valuable insights into the distribution of data, helping to understand the characteristics and patterns within the dataset. Skewness and kurtosis are important tools in statistical analysis for assessing the shape of data distributions and identifying any deviations from normality.